Problem

Two loads connected in parallel are respectively 3 kW at a pf of 0.75 leading and 6 kW at a pf of 0.95 lagging. Calculate the pf of the combined two loads. Find the complex power supplied by the source.

Step-by-step solution

Step 1 of 9

Consider the circuit diagram shown in Figure 1.

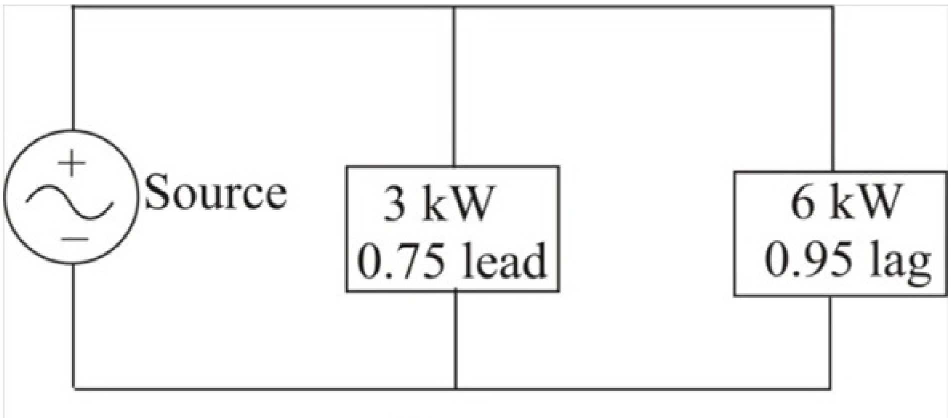

Figure 1

Step 2 of 9

Consider the expression for the reactive power Q.

$$Q = S\sin\theta$$

Substitute $\frac{P}{\cos\theta}$ for S.

$$\begin{aligned} Q &= \left(\frac{P}{\cos\theta}\right)\sin\theta \\ &= P\tan\theta \end{aligned}$$

Therefore, the expression for the reactive power Q is, $P\tan\theta$.

Step 3 of 9

Calculate the power factor angle θ_1 at load 1.

$$\begin{aligned} \theta_1 &= \cos^{-1}(0.75) \\ &= 41.41° \end{aligned}$$

Therefore, the power factor angle θ_1 at load 1 is, $41.41°$.

Step 4 of 9

Calculate the power factor angle θ_2 at load 2.

$$\theta_2 = \cos^{-1}(0.95)$$
$$= 18.2°$$

Therefore, the power factor angle θ_2 at load 2 is, $18.2°$.

Step 5 of 9

Calculate the value of the reactive power at load 1.

$$Q_1 = P_1 \tan\theta_1$$

Substitute 3 kW for P_1 and $41.41°$ for θ_1. Notice that reactive power is negative since power factor is leading.

$$Q_1 = -(3\text{ k})\tan(41.41°)$$
$$= -(3\text{ k})(0.8819)$$
$$= -2.6457\text{ kVAR}$$

Therefore, the value of the reactive power Q_1 at load 1 is, 2.6457 kVAR.

Step 6 of 9

Calculate the value of the reactive power at load 2.

$$Q_2 = P_2 \tan\theta_2$$

Substitute 6 kW for P_2 and $18.2°$ for θ_2.

$$Q_2 = (6\text{ k})\tan(18.2°)$$
$$= (6\text{ k})(0.3288)$$
$$= 1.9728\text{ kVAR}$$

Therefore, the value of the reactive power Q_2 at load 2 is, 1.9728 kVAR.

Step 7 of 9

Calculate the value of the total complex power S_T absorbed by the two loads.

$$\text{S}_\text{T} = \text{S}_1 + \text{S}_2$$

Substitute $P_1 + jQ_1$ for S_1 and $P_2 + jQ_2$ for S_2.

$$\text{S}_\text{T} = P_1 + jQ_1 + P_2 + jQ_2$$

Substitute 3 kW for P_1, -2.6457 kVAR for Q_1, 6 kW for P_2 and 1.9728 kVAR for Q_2.

$$\text{S}_\text{T} = 3\text{ k} - j2.6457\text{ k} + 6\text{ k} + j1.9728\text{ k}$$
$$= 9\text{ k} - j0.6729\text{ k}$$
$$= (9 - j0.6729)\text{ kVA}$$

Therefore, the value of the total complex power absorbed by the two loads is,

$(9 - j0.6729)\text{ kVA}$.

The total power absorbed by the two loads is equal to the power supplied by the source. That is, $\boxed{(9 - j0.6729)\text{ kVA}}$.

The value of the total real power P_T and the total reactive power Q_T absorbed by the load is, 9 kW and 0.6729 kVAR respectively.

Step 8 of 9

Calculate the value of the apparent power absorbed by the two loads.

$$S_T = |\mathrm{S_T}|$$

Substitute $(9 - j0.6729)\ \mathrm{kVA}$ for $\mathrm{S_T}$.

$$\begin{aligned} S_T &= |(9 - j0.6729)\ \mathrm{k}| \\ &= \sqrt{(9\ \mathrm{k})^2 + (0.6729\ \mathrm{k})^2} \\ &= \sqrt{81 + 0.4537}\ \mathrm{k} \\ &= \sqrt{81.4537}\ \mathrm{k} \end{aligned}$$

$$S_T = 9.025\ \mathrm{kVA}$$

Therefore, the value of the apparent power S_T absorbed by the two loads is, $9.025\ \mathrm{kVA}$.

Step 9 of 9

Calculate the combined power factor $\mathrm{pf_T}$ of the two loads.

$$\mathrm{pf_T} = \frac{P_T}{S_T}$$

Substitute $9\ \mathrm{kW}$ for P_T and $9.025\ \mathrm{kVA}$ for S_T.

$$\begin{aligned} \mathrm{pf_T} &= \frac{9\ \mathrm{k}}{9.025\ \mathrm{k}} \\ &= 0.9972\ \text{leading} \end{aligned}$$

Power factor is leading since the value of the reactive power is negative.

Therefore, the value of the combined power factor $\mathrm{pf_T}$ is, $\boxed{0.9972\ \text{leading}}$.